ECE 4502/6502 & CS 6501: Machine Learning on Graphs

Lecture 4. Network Measures (II)

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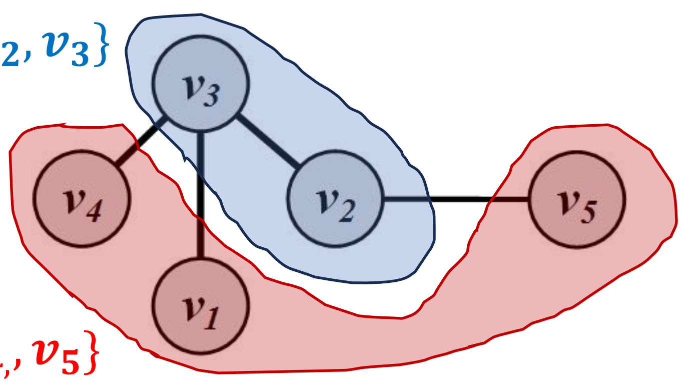
Centrality for a group of nodes

Group Centrality

- How to measure the centrality of a group of nodes?
- Key idea:
    - Generalize node centrality to group centrality
- Approach
    - Combine all nodes in a group with a super node

Let S denote the set of nodes in the group and $V - S$ the set of outsiders

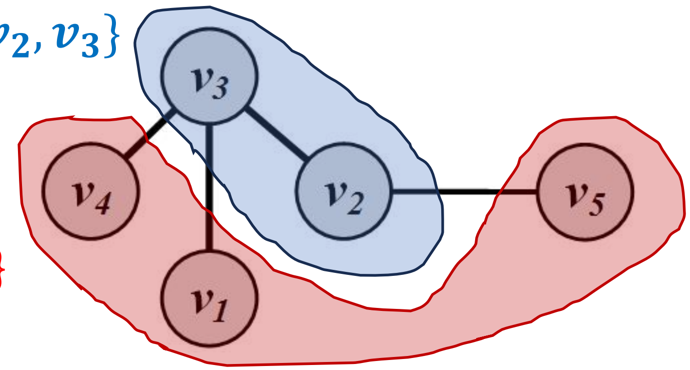
$S = \{v_2, v_3\}$



$V - S = \{v_1, v_{4,}, v_5\}$

Group Centrality

$S = \{v_2, v_3\}$



$V - S = \{v_1, v_4, v_5\}$

I. Group Degree Centrality

$$C_d^{\text{group}}(S) = |\{v_i \in V - S | v_i \text{ is connected to } v_j \in S\}|$$

- **Normalization**: divide by $|V - S|$

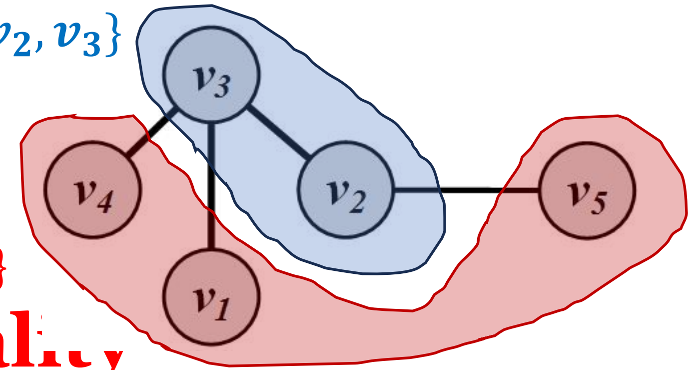
II. Group Betweenness Centrality

$$C_b^{\text{group}}(S) = \sum_{s \neq t, s \notin S, t \notin S} \frac{\sigma_{st}(S)}{\sigma_{st}}$$

- **Normalization**: divide by $2\binom{|V-S|}{2}$

Group Centrality

$S = \{v_2, v_3\}$



$V - S = \{v_1, v_4, v_5\}$

III. Group Closeness Centrality

$$C_c^{\text{group}}(S) = \frac{1}{\bar{l}_S^{\text{group}}}$$

- It is the average distance from non-members to the group

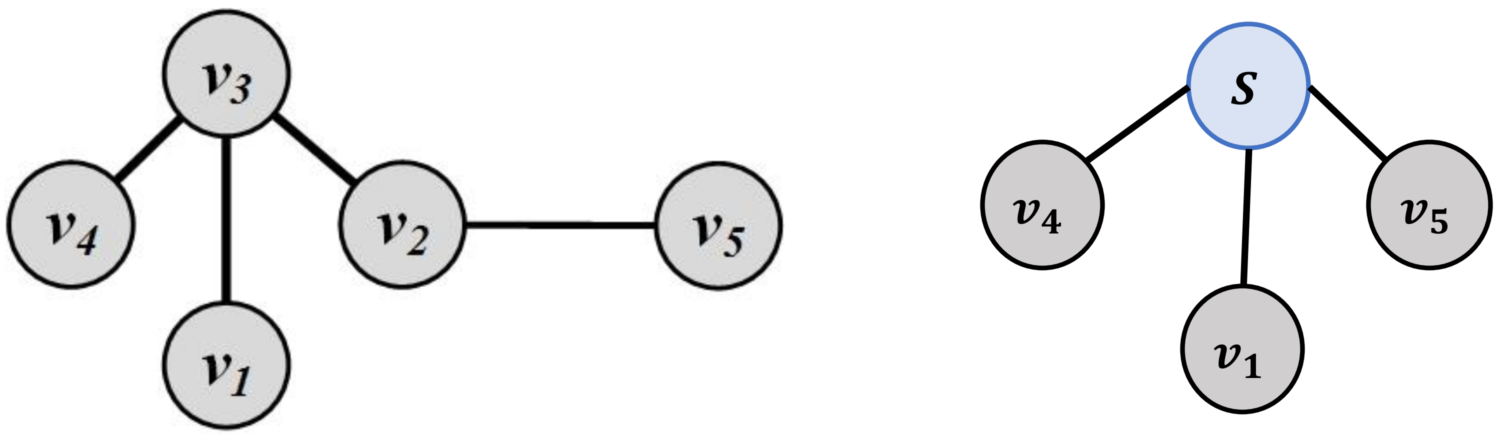
$$\bar{l}_S^{\text{group}} = \frac{1}{|V-S|} \sum_{v_j \notin S} l_{S,v_j}$$

$$l_{S,v_j} = \min_{v_i \in S} l_{v_i,v_j}$$

- One can also utilize the *maximum distance* or the *average distance*

Group Centrality Example

- Consider $S = \{v_2, v_3\}$



- Group degree centrality = **3**
- Group betweenness centrality = **6**
- Group closeness centrality = **1**

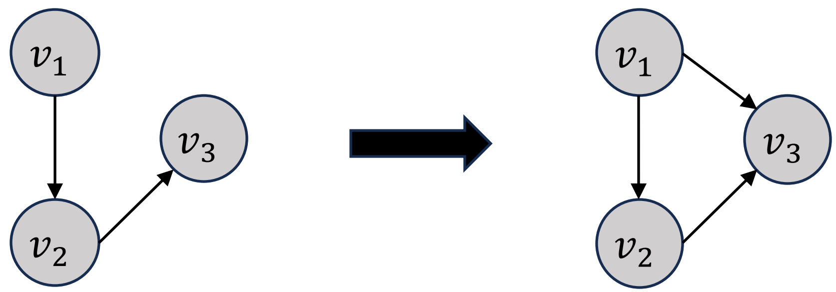
Connection Patterns

- Transitivity/Reciprocity
- Status/Balance

Transitivity and Reciprocity

Transitivity

- Mathematic representation:
  - For a transitive relation R: $aRb \wedge bRc \rightarrow aRc$

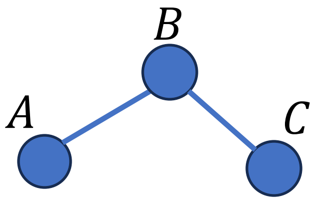


- In a social network:
  - ***Transitivity is when a friend of my friend is my friend***
  - Transitivity in a social network leads to a denser graph, which in turn is closer to a complete graph
  - We can determine how close graphs are to the complete graph by measuring transitivity

[Global] Clustering Coefficient

- **Clustering coefficient** measures transitivity in undirected graphs
  - Count paths of length two and check whether the third edge exists

$$C = \frac{|\text{Closed Paths of Length 2}|}{|\text{Paths of Length 2}|}$$



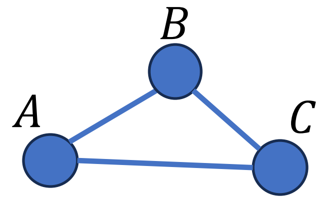
Paths of length 2:

ABC, CBA

Closed paths of length 2:

None

Clustering coefficient $C = 0$



Paths of length 2:

$ABC, CBA, BAC, CAB, BCA, ACB$

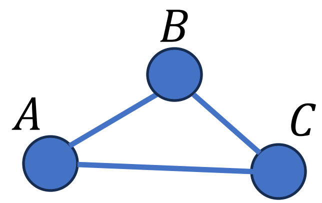
Closed paths of length 2:

$ABC, CBA, BAC, CAB, BCA, ACB$

Clustering coefficient $C = 1$

[Global] Clustering Coefficient

- **Clustering coefficient**
  - Count with triangles
    - 1 triangle = 6 closed paths of length 2

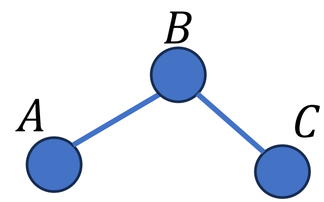


Closed paths of length 2:
$ABC, CBA, BAC, CAB, BCA, ACB$

Triples:
$\{ABC\}, \{BCA\}, \{CAB\}$

$$C = \frac{(\text{Number of Triangles}) \times 6}{|\text{Paths of Length } 2|}$$

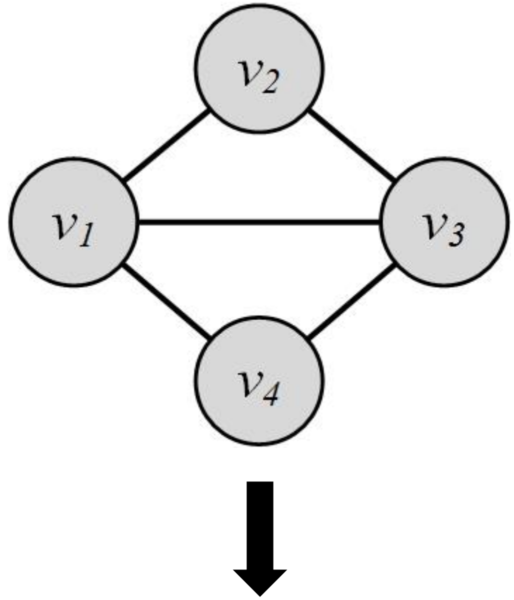
- Count with connected triples
  - Triple = an ordered set of three nodes
  - 1 connected triple = 2 paths of length 2



$triple\ \{ABC\} = triple\ \{CBA\}$

$$C = \frac{(\text{Number of Triangles}) \times 3}{\text{Number of Connected Triples of Nodes}}$$

[Global] Clustering Coefficient: Example



$$
\begin{aligned}
C &= \frac{(\text{Number of Triangles}) \times 3}{\text{Number of Connected Triples of Nodes}} \\
&= \frac{2 \times 3}{2 \times 3 + \underbrace{2}_{v_2 v_1 v_4,\, v_2 v_3 v_4}} = 0.75.
\end{aligned}
$$

[Local] Clustering Coefficient

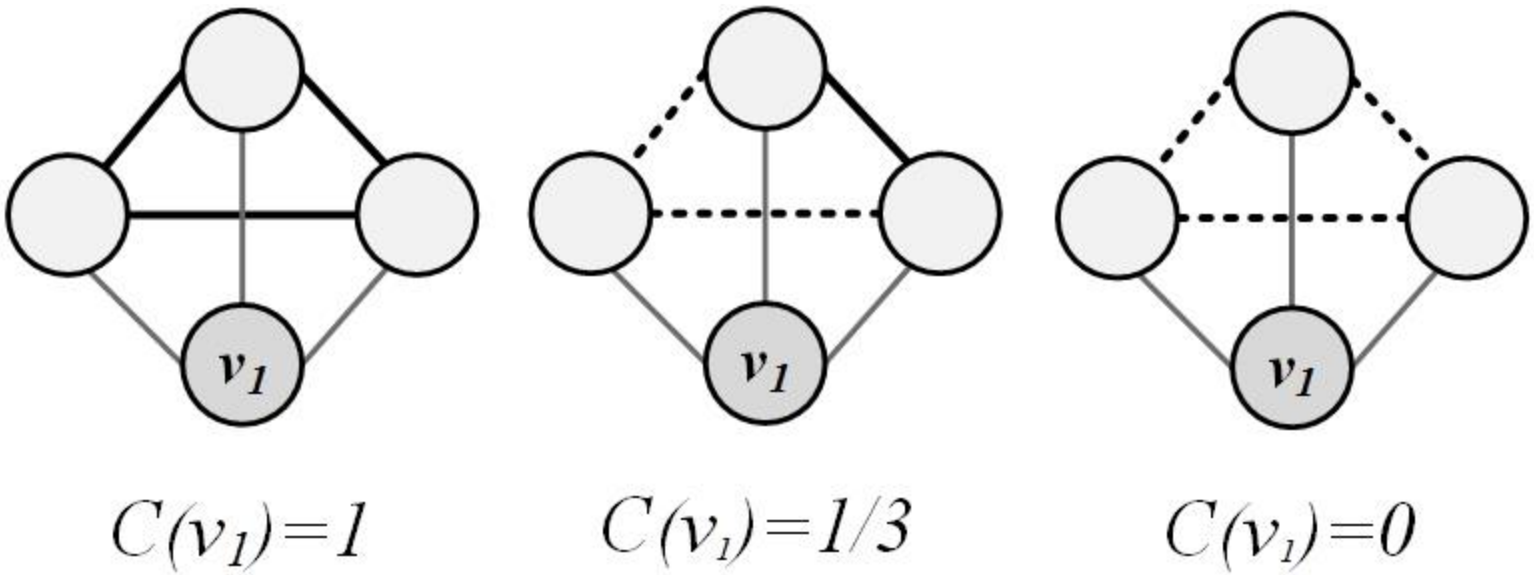
- Local clustering coefficient measures transitivity at the **node level**
  - Commonly employed for undirected graphs
  - Computes how strongly the neighbors of a node v are connected

$$C(v_i) = \frac{\text{Number of Pairs of Neighbors of } v_i \text{ That Are Connected}}{\text{Number of Pairs of Neighbors of } v_i}.$$

In an undirected graph, the number of pairs of v_i's neighbors (i.e., the denominator) can be rewritten as:

$$\binom{d_i}{2} = d_i(d_i - 1)/2$$

[Local] Clustering Coefficient: Example



- Thin lines depict connections to neighbors
- Dashed lines are the missing link among neighbors
- Solid lines indicate connected neighbors
  - When none of neighbors are connected $C = 0$
  - When all neighbors are connected $C = 1$

Clustering Coefficient

- In real-world networks, friendships are highly transitive

Facebook

May 2011:
Average clustering coefficient of **0.5** for users with **two** friends

  - Friends of a user are often friends with one another
  - These friendships form triads
  - High average [local] clustering coefficient

Web	Facebook	Flickr	LiveJournal	Orkut	YouTube
0.081	0.14 (with 100 friends)	0.31	0.33	0.17	0.13

Clustering Coefficient for Real-World Networks

	Network	Type	n	m	C
Social	Film actors	Undirected	449 913	25 516 482	0.20
	Company directors	Undirected	7 673	55 392	0.59
	Math coauthorship	Undirected	253 339	496 489	0.15
	Physics coauthorship	Undirected	52 909	245 300	0.45
	Biology coauthorship	Undirected	1 520 251	11 803 064	0.088
	Telephone call graph	Undirected	47 000 000	80 000 000	
	Email messages	Directed	59 812	86 300	
	Email address books	Directed	16 881	57 029	0.17
	Student dating	Undirected	573	477	0.005
	Sexual contacts	Undirected	2 810		
Information	WWW `nd.edu`	Directed	269 504	1 497 135	0.11
	WWW AltaVista	Directed	203 549 046	1 466 000 000	
	Citation network	Directed	783 339	6 716 198	
	Roget's Thesaurus	Directed	1 022	5 103	0.13
	Word co-occurrence	Undirected	460 902	16 100 000	
Technological	Internet	Undirected	10 697	31 992	0.035
	Power grid	Undirected	4 941	6 594	0.10
	Train routes	Undirected	587	19 603	
	Software packages	Directed	1 439	1 723	0.070
	Software classes	Directed	1 376	2 213	0.033
	Electronic circuits	Undirected	24 097	53 248	0.010
	Peer-to-peer network	Undirected	880	1 296	0.012
Biological	Metabolic network	Undirected	765	3 686	0.090
	Protein interactions	Undirected	2 115	2 240	0.072
	Marine food web	Directed	134	598	0.16
	Freshwater food web	Directed	92	997	0.20
	Neural network	Directed	307	2 359	0.18

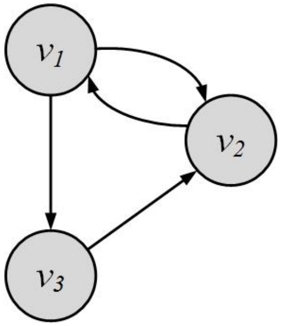
Reciprocity

If you follow me, I'll follow back

- Reciprocity is simplified version of transitivity
  - It considers closed loops of length 2

$$R = \frac{|\text{Closed loops of length 2}|}{\text{max possible closed loops of length 2}}$$

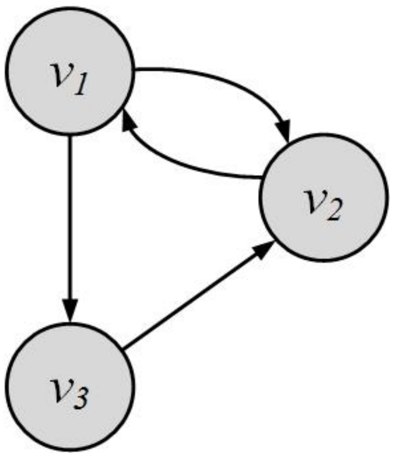
$$= \frac{|\text{Closed loops of length 2}|}{|E|/2}$$



$$\begin{aligned} R &= \frac{\sum_{i,j,i<j} A_{i,j} A_{j,i}}{|E|/2}, \\ &= \frac{2}{|E|} \sum_{i,j,i<j} A_{i,j} A_{j,i}, \\ &= \frac{2}{|E|} \times \frac{1}{2} \mathrm{Tr}(A^2), \\ &= \frac{1}{|E|} \mathrm{Tr}(A^2), \\ &= \frac{1}{m} \mathrm{Tr}(A^2). \end{aligned}$$

$$\boxed{\mathrm{Tr}(A) = A_{1,1} + A_{2,2} + \cdots + A_{n,n} = \sum_{i=1}^{n} A_{i,i}}$$

Reciprocity: Example



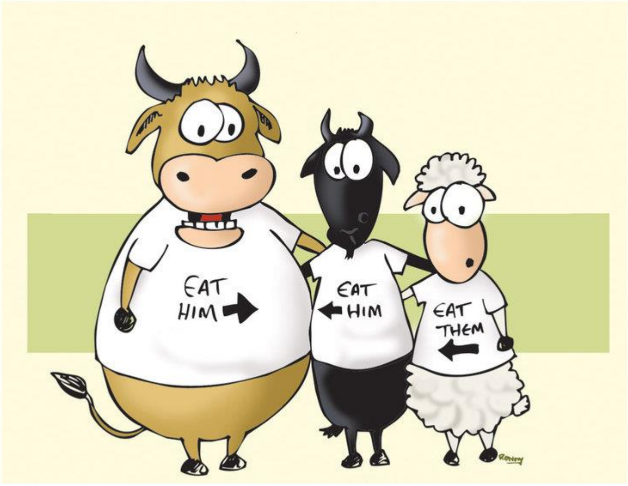
$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Reciprocal nodes: v_1, v_2

$$R = \frac{1}{m}\mathrm{Tr}(A^2) = \frac{1}{4}\mathrm{Tr}\left(\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}\right) = \frac{2}{4} = \frac{1}{2}.$$

Balance and Status

• Measuring consistency in friendships



Social Balance Theory

Social balance theory

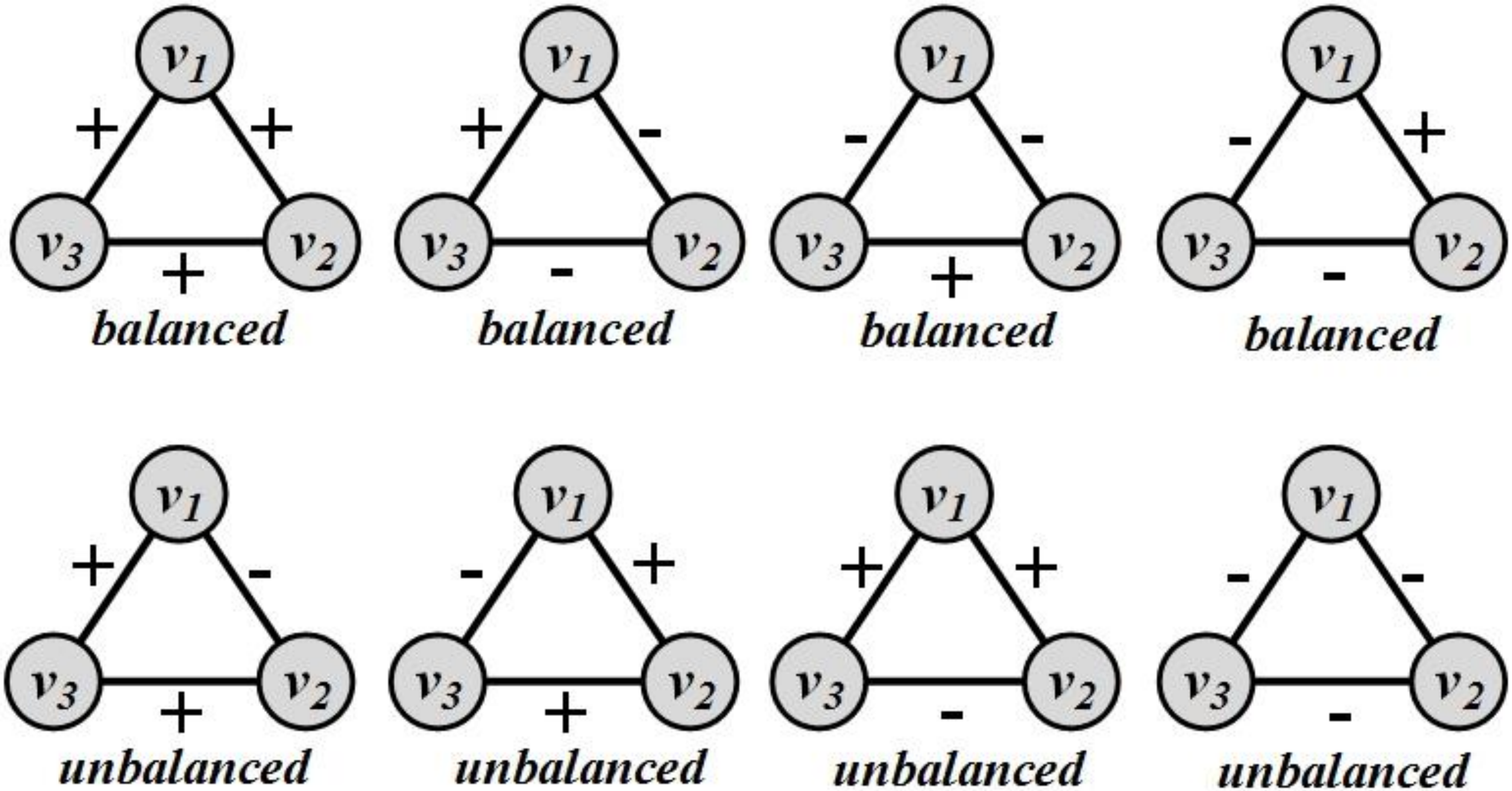
- Consistency in friend/foe relationships among individuals
- Informally, friend/foe relationships are consistent when

The friend of my friend is my friend,
The friend of my enemy is my enemy,
The enemy of my enemy is my friend,
The enemy of my friend is my enemy.

- In the network
  - Positive edges demonstrate friendships ($w_{ij} = 1$)
  - Negative edges demonstrate being enemies ($w_{ij} = -1$)
- Triangle of nodes i, j, and k, is balanced, if and only if
  - w_{ij} denotes the value of the edge between nodes i and j

$$w_{ij} w_{jk} w_{ki} \geq 0.$$

Social Balance Theory: Possible Combinations



For any cycle, if the multiplication of edge values become positive, then the cycle is socially balanced

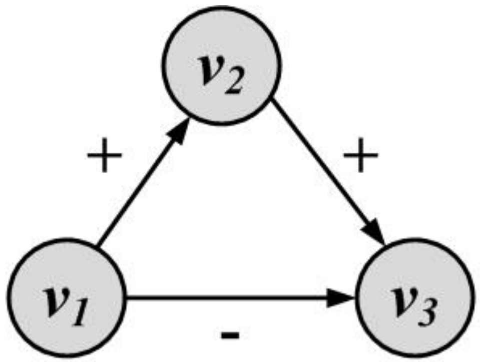
Social Status Theory

- **Status:** how prestigious an individual is ranked within a society

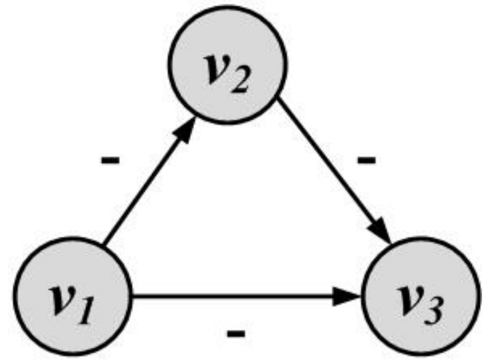
- **Social status theory:**
  - How consistent individuals are in assigning status to their neighbors
  - Informally,

If X has a higher status than Y and Y has a higher status than Z, then X should have a higher status than Z.

Social Status Theory: Example



Unstable configuration



Stable configuration

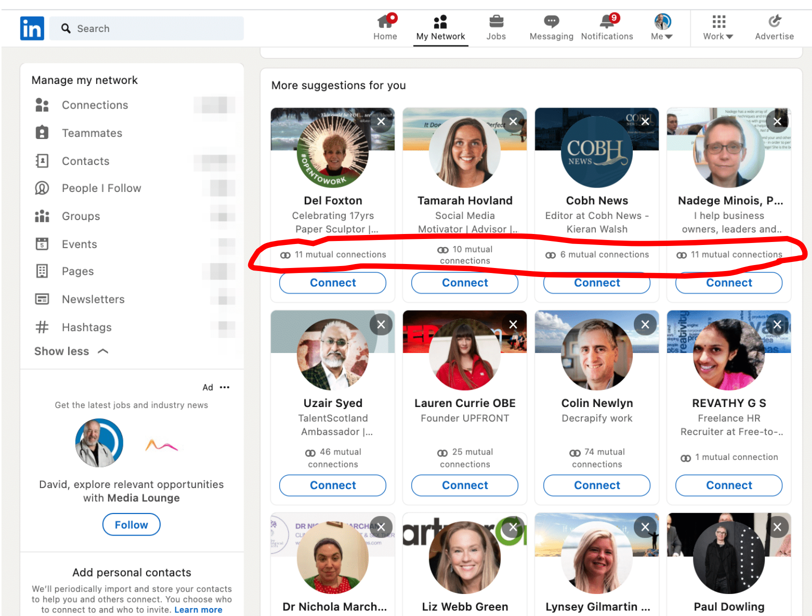
- A directed '+' edge from node *X* to node *Y* shows that *Y* has a higher status than *X* and a '-' one shows vice versa

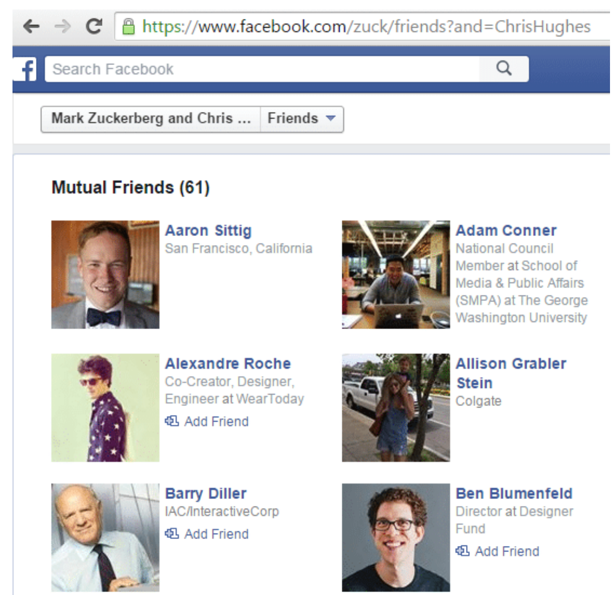
Similarity

- Structural Equivalence
- Regular Equivalence

Structural Equivalence

- **Structural Equivalence:**
  - We look at the neighborhood <u>shared</u> by two nodes;
  - The size of this shared neighborhood defines how similar two nodes are.
- ***Example:*** Mutual connections/friends





Structural Equivalence: Definitions

- **Vertex similarity**: $\sigma(v_i, v_j) = |N(v_i) \cap N(v_j)|$

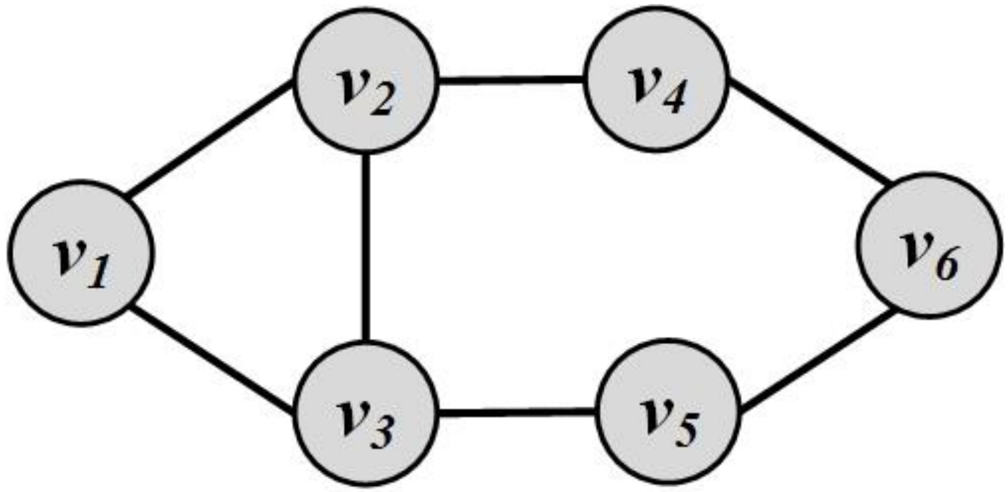
Normalize with neighbor size

Jaccard Similarity: $\sigma_{Jaccard}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{|N(v_i) \cup N(v_j)|}$

Cosine Similarity: $\sigma_{Cosine}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{\sqrt{|N(v_i)||N(v_j)|}}$

- The neighborhood $N(v)$ often excludes the node itself v.
  - **What can go wrong?**
    - Connected nodes not sharing a neighbor will be assigned zero similarity
  - **Solution:**
    - We can assume nodes are included in their neighborhoods

Similarity: Example



$$\sigma_{\text{Jaccard}}(v_2, v_5) = \frac{|\{v_1, v_3, v_4\} \cap \{v_3, v_6\}|}{|\{v_1, v_3, v_4, v_6\}|} = 0.25$$

$$\sigma_{Cosine}(v_2, v_5) = \frac{|\{v_1, v_3, v_4\} \cap \{v_3, v_6\}|}{\sqrt{|v_1, v_3, v_4||v_3, v_6|}} = \frac{1}{\sqrt{6}} \approx 0.41$$

Similarity Significance

Measuring Similarity Significance: compare the calculated similarity value with its expected value where vertices pick their neighbors at random

- For vertices v_i and v_j with degrees d_i and d_j this expectation is $d_i d_j / n$
  - There is a d_i / n chance of becoming v_i's neighbor
  - v_j selects d_j neighbors
- We can rewrite neighborhood overlap as

$$\sigma(v_i, v_j) = |N(v_i) \cap N(v_j)| = \sum_k A_{i,k} A_{j,k}$$

Normalized Similarity Cont.

$$\bar{A}_i = \frac{1}{n}\sum_k A_{i,k}$$

$$\begin{aligned}
\sigma_{\text{significance}}(v_i, v_j) &= \sum_k A_{i,k}A_{j,k} - \frac{d_i d_j}{n} \\
&= \sum_k A_{i,k}A_{j,k} - n\frac{1}{n}\sum_k A_{i,k}\frac{1}{n}\sum_k A_{j,k} \\
&= \sum_k A_{i,k}A_{j,k} - n\bar{A}_i\bar{A}_j \\
&= \sum_k (A_{i,k}A_{j,k} - \bar{A}_i\bar{A}_j) \\
&= \sum_k (A_{i,k}A_{j,k} - \bar{A}_i\bar{A}_j - \bar{A}_i\bar{A}_j + \bar{A}_i\bar{A}_j) \\
&= \sum_k (A_{i,k}A_{j,k} - A_{i,k}\bar{A}_j - \bar{A}_iA_{j,k} + \bar{A}_i\bar{A}_j) \\
&= \boxed{\sum_k (A_{i,k} - \bar{A}_i)(A_{j,k} - \bar{A}_j)}
\end{aligned}$$

What is this?

Normalized Similarity Cont.

***n* times the Covariance between A_i and A_j**

$$\frac{1}{n}\sum_k (A_{i,k} - \bar{A}_i)(A_{j,k} - \bar{A}_j)$$

Normalize covariance by the multiplication of Variances.

$$\sqrt{\frac{1}{n}\sum_k (A_{i,k} - \bar{A}_i)^2}\sqrt{\frac{1}{n}\sum_k (A_{j,k} - \bar{A}_j)^2}$$

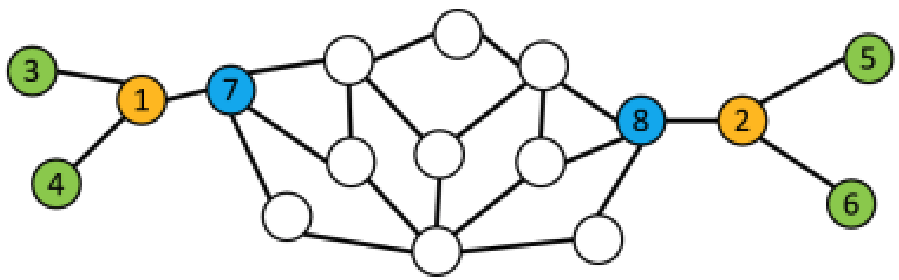
We get **Pearson correlation coefficient**

$$\begin{aligned}\sigma_{\text{pearson}}(v_i, v_j) &= \frac{\sigma_{\text{significance}}(v_i, v_j)}{\sqrt{\sum_k (A_{i,k} - \bar{A}_i)^2}\sqrt{\sum_k (A_{j,k} - \bar{A}_j)^2}} \\ &= \frac{\sum_k (A_{i,k} - \bar{A}_i)(A_{j,k} - \bar{A}_j),}{\sqrt{\sum_k (A_{i,k} - \bar{A}_i)^2}\sqrt{\sum_k (A_{j,k} - \bar{A}_j)^2}}\end{aligned}$$

(range of $\sigma \in [-1,1]$)

Regular Equivalence

- Regular equivalence,
  - Two vertexes are regularly equivalent if their **neighbor structures** are similar

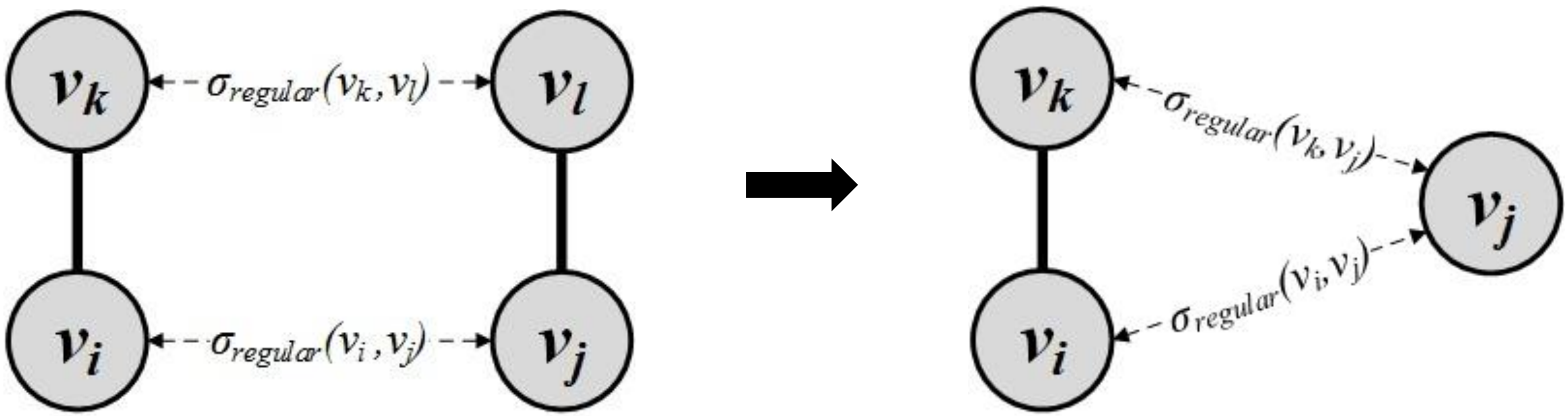


- ***Example***:
  - Nodes with the same color are regularly equivalent
    Husband – Wife – Kids

Regular Equivalence

- v_i, v_j are similar when their neighbors v_k and v_l are similar

$$\sigma_{\text{regular}}(v_i, v_j) = \alpha \sum_{k,l} A_{i,k} A_{j,l} \sigma_{\text{regular}}(v_k, v_l)$$



- The equation (left figure) is hard to solve since it is self-referential so we relax our definition using the right figure

Regular Equivalence

- v_i and v_j are similar when v_j is similar to v_i's neighbors v_k

$$\sigma_{regular}(v_i, v_j) = \alpha \sum_k A_{i,k} \sigma_{Regular}(v_k, v_j)$$

- In vector format

$$\sigma_{regular} = \alpha A \sigma_{Regular}$$

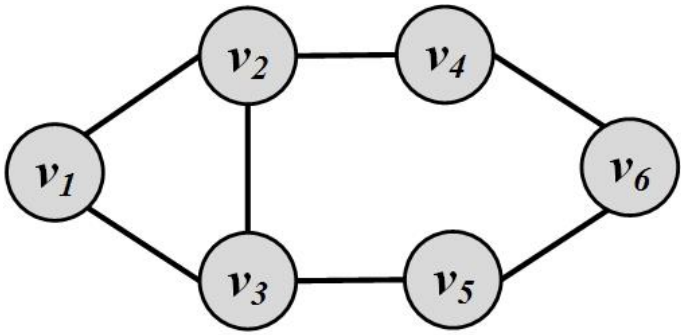
<u>A vertex is highly similar to itself</u>, we guarantee this by adding an identity matrix to the equation

$$\sigma_{regular} = \alpha A \sigma_{Regular} + \mathbf{I}$$

$$\sigma_{regular} = (\mathbf{I} - \alpha A)^{-1}$$

When $\alpha < 1/\lambda_{max}$ the matrix is invertible

Regular Equivalence: Example



$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

The largest eigenvalue of A is 2.43

Set $\alpha = 0.3 < 1/2.43$

$$\sigma_{\text{regular}} = (I - 0.3A)^{-1} = \begin{bmatrix} 1.43 & 0.73 & 0.73 & 0.26 & 0.26 & 0.16 \\ 0.73 & 1.63 & 0.80 & 0.56 & 0.32 & 0.26 \\ 0.73 & 0.80 & 1.63 & 0.32 & 0.56 & 0.26 \\ 0.26 & 0.56 & 0.32 & 1.31 & 0.23 & 0.46 \\ 0.26 & 0.32 & 0.56 & 0.23 & 1.31 & 0.46 \\ 0.16 & 0.26 & 0.26 & 0.46 & 0.46 & 1.27 \end{bmatrix}$$

- Any row/column of this matrix shows the similarity to other vertices
- Vertex 1 is most similar (other than itself) to vertices 2 and 3
- Nodes 2 and 3 have the highest similarity (**regular equivalence**)

Questions?